

AdaDexGrasp: Adaptive Dexterous Grasping via 3D Visuo-Tactile Representation Fusion

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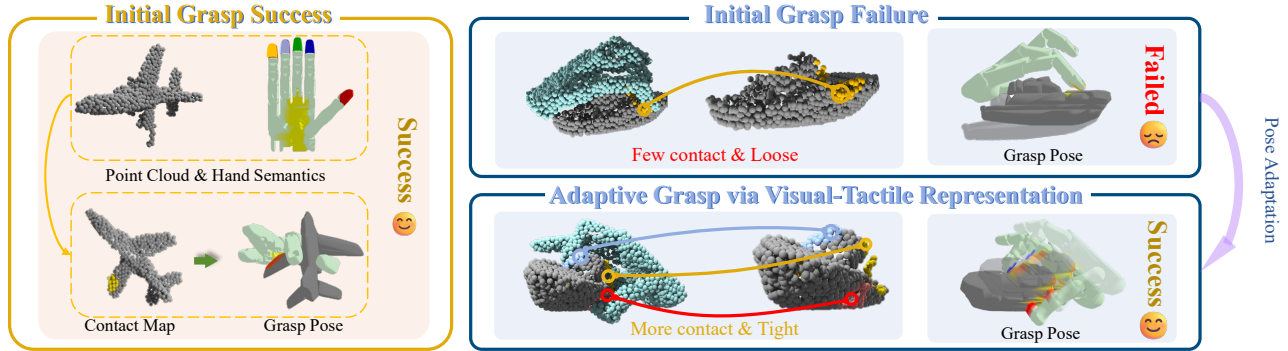


Fig. 1. **Overview.** (Left) An initial grasping module predicts a contact map from the raw point cloud with finger/palm identities, aiming to generate physically successful hand poses for grasp planning informed by hand semantic information. (Right-Top) The initially generated hand pose does not guarantee a successful grasp and may fail due to sparse or unstable contact formations. (Right-Bottom) Our method further leverages a visuo-tactile-fusion representation to adapt the initial pose, resulting in more stable contact patterns that enable robust grasp execution.

Abstract—Humans achieve stable and adaptive grasps by seamlessly integrating visual perception and tactile feedback, a capability that remains challenging to replicate in robotic systems. Existing robotic grasping approaches predominantly rely on visual inputs and lack mechanisms for tactile-guided adaptation after contact, limiting robustness and generalization. To address this challenge, we propose a unified visuo-tactile-fusion grasping framework that integrates grasp generation, feasibility prediction, and adaptive refinement. At its core, our method introduces an efficient visuo-tactile representation that tightly fuses object geometry with tactile feedback by associating tactile signals with finger identities. This unified representation supports contact-aware grasp pose generation during planning and tactile-guided refinement after contact, enabling the system to reason about fine-grained finger-object interactions and adjust grasps dynamically. Comprehensive experiments in both simulation and real-world environments demonstrate that our approach significantly enhances grasp success rates and generalization across diverse objects.

I. INTRODUCTION

Humans achieve robust dexterous grasping by combining visual perception for grasp planning with tactile feedback for real-time adjustment. Transferring visuo-tactile grasping capabilities to dexterous hands is important yet challenging.

Most existing dexterous grasping methods [1, 2, 3, 4, 5, 6, 7, 8] rely primarily on visual observations to generate grasp poses, without adapting the grasp based on feedback after contact. Tactile-based approaches [9, 10] enable reactive adjustment but often underutilize available visual information. Recent visuo-tactile methods [11, 12, 13, 14, 15] integrate both modalities, yet mostly focus on feature-level fusion or manipulation tasks, while the spatial correspondence between tactile contacts and object geometry remains underexplored for generalizable dexterous grasping.

To address these limitations, we propose a unified visuo-tactile framework that integrates tactile feedback with object

geometry for grasp generation, feasibility prediction, and adaptive refinement. Our key idea is to explicitly associate tactile contacts with object geometry, enabling fine-grained reasoning about where different parts of the hand should interact with the object.

For initial grasp generation, we introduce a *semantic contact map* that associates object contact regions with specific finger/palm identities. Unlike conventional contact representations that only encode whether contact occurs [16, 17, 18], our representation captures explicit hand-object contact correspondences. Incorporating this representation into object point-cloud allows the model to generate grasp poses that anticipate desirable tactile contacts.

For challenging or unseen objects, the initial grasp may still be unstable [19, 20]. We therefore further exploit tactile feedback during execution. Tactile signals are mapped onto the current hand-object geometry as contact markers indicating different finger/palm contacts. Based on this tactile-mapped geometric representation, a classifier predicts grasp feasibility, while an adaptation model refines unsuccessful grasps toward more stable configurations.

To support efficient learning and evaluation, we further develop a tactile-enabled dexterous manipulation environment in IsaacGym [21] and deploy our framework on a real robotic system. Extensive simulation and real-world experiments demonstrate strong generalization and robustness across diverse object geometries.

II. METHOD

A. Overview

To address this challenge, we propose a unified framework that integrates object geometry and tactile feedback to support grasp pose generation, feasibility prediction, and

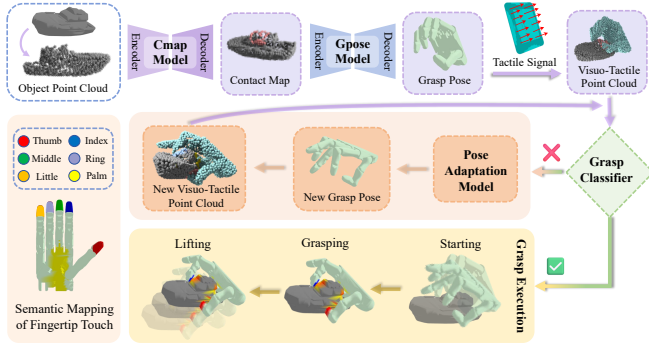


Fig. 2. **Pipeline of the proposed visuo-tactile grasping framework.** Given an object point cloud and semantic fingertip labels, the system first predicts a contact map using the Cmap model and generates an initial grasp pose with the Gpose model. The predicted hand configuration and tactile feedback are fused into a visuo-tactile point cloud that captures both geometric and contact cues. A grasp classifier evaluates grasp stability, and if failure is predicted, a pose adaptation module iteratively refines the grasp by generating updated poses and visuo-tactile observations until success is predicted or a maximum number of iterations is reached. The final grasp is executed through a staged process of reaching, grasping, and lifting, enabling stable interaction with diverse objects.

adaptive refinement. The overall pipeline is illustrated in Figure 2. The architecture of the proposed method (Figure 3) consists of three components:

- **Contact-Driven Grasp Pose Generator** (Section II-B), which produces the initial grasp pose.
- **Grasp Pose Classifier Model** (Section II-C), which predicts grasp feasibility and stability.
- **Grasp Pose Adaptation Model** (Section II-D), which refines suboptimal grasp poses.

Closed-Loop Grasp Optimization (Section II-E) integrates these components into an iterative refinement process.

B. Contact-Driven Grasp Pose Generator

Given object point cloud, we generate an initial grasp hypothesis by predicting a contact distribution over the object surface. Model h_ψ predicts a probabilistic contact map $\mathcal{M}$:

$$\mathcal{M} = \{\mathcal{M}_i \mid \mathcal{M}_i \in \{1, 2, 3, 4, 5, 6\}, i = 1, 2, \dots, N_0\} = h_{\psi_1}(\mathcal{P}_{obj}) \quad (1)$$

where each label $\mathcal{M}_i$ corresponds to a specific hand part (finger or palm). Thus, the predicted contact map encodes potential tactile interactions between the hand and the object.

The grasp generator h'_{ψ} then conditions on $(\mathcal{P}_{obj}, \mathcal{M})$ to predict an initial grasp state:

$$s^{(0)} = h'_{\psi_2}(\mathcal{P}_{obj}, \mathcal{M}). \quad (2)$$

This estimate s_0 serves as an initial grasp pose that often succeeds directly or provides a good starting point for subsequent refinement.

During initial grasp generation, tactile inputs are unavailable because no contact has occurred. Each training sample includes hand joint states, grasp success labels, and visuo-tactile point clouds. The contact map prediction is supervised using cross-entropy loss:

$$\mathcal{L}_{CE}(\mathcal{M}, \mathcal{M}_s) = - \sum_i M_{s,i} \log M_i \quad (3)$$

while the grasp pose is supervised using MSE:

$$\mathcal{L}_{MSE} = \|s^{(0)} - s^s\|_2^2 \quad (4)$$

Training data are collected through RL rollouts and can be adapted to different dexterous hands by training the policy on the corresponding hand models.

C. Grasp Pose Classifier Model

Due to the diversity of hand poses and object-dependent grasp requirements, an initially generated grasp may not always succeed. We therefore predict grasp feasibility based on the current hand-object interaction state.

After executing the initial grasp pose s_0 , we annotate the object point cloud with tactile signals derived from contact forces at the five fingertips and the palm. Each contact region is assigned a unique ID corresponding to the contacting hand part, producing a tactile-mapped point cloud $\mathcal{P}_{vt}$.

Given $\mathcal{P}_{vt}$, the classifier estimates the probability that the current configuration yields a stable grasp. Formally, the model learns a discriminative mapping:

$$\hat{y} = f_\theta(\mathcal{P}_{vt}, s), \quad (5)$$

where $\hat{y} \in [0, 1]$ denotes the predicted grasp success probability and f_θ is a PointNet-based network. The model is trained using binary cross-entropy:

$$\mathcal{L}_{cls} = -[y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})]. \quad (6)$$

By leveraging both geometric structure and tactile contact patterns, f_θ provides a differentiable estimate of grasp stability that can be evaluated in real time.

D. Grasp Pose Adaptation Model

To recover from failed grasps, we introduce a grasp pose adaptation model g_ϕ that transforms unstable grasps into stable ones.

For training, we construct paired tactile-enhanced states:

$$(\mathcal{P}_{vt}^f, s^f, y = 0), \quad (\mathcal{P}_{vt}^s, s^s, y = 1),$$

where the successful grasp s^s is geometrically and temporally close to the failed grasp s^f . Concretely, for each object we pool successful and failed states from the PPO rollouts immediately before lifting, so that all candidates share the same interaction stage, and transform them into an object-centric frame to remove any dependence on absolute object placement. Each failed state s^f is then paired with its nearest successful counterpart s^s under the distance

$$d(s, s') = 10 \sqrt{\|p - p'\|_2^2 + \theta_{q,q'}^2} + \|\mathbf{j} - \mathbf{j}'\|_2, \quad (7)$$

where $(p, q, \mathbf{j})$ denote the palm translation, palm orientation, and the 22 joint angles, and $\theta_{q,q'}$ is the geodesic distance between the two orientations.

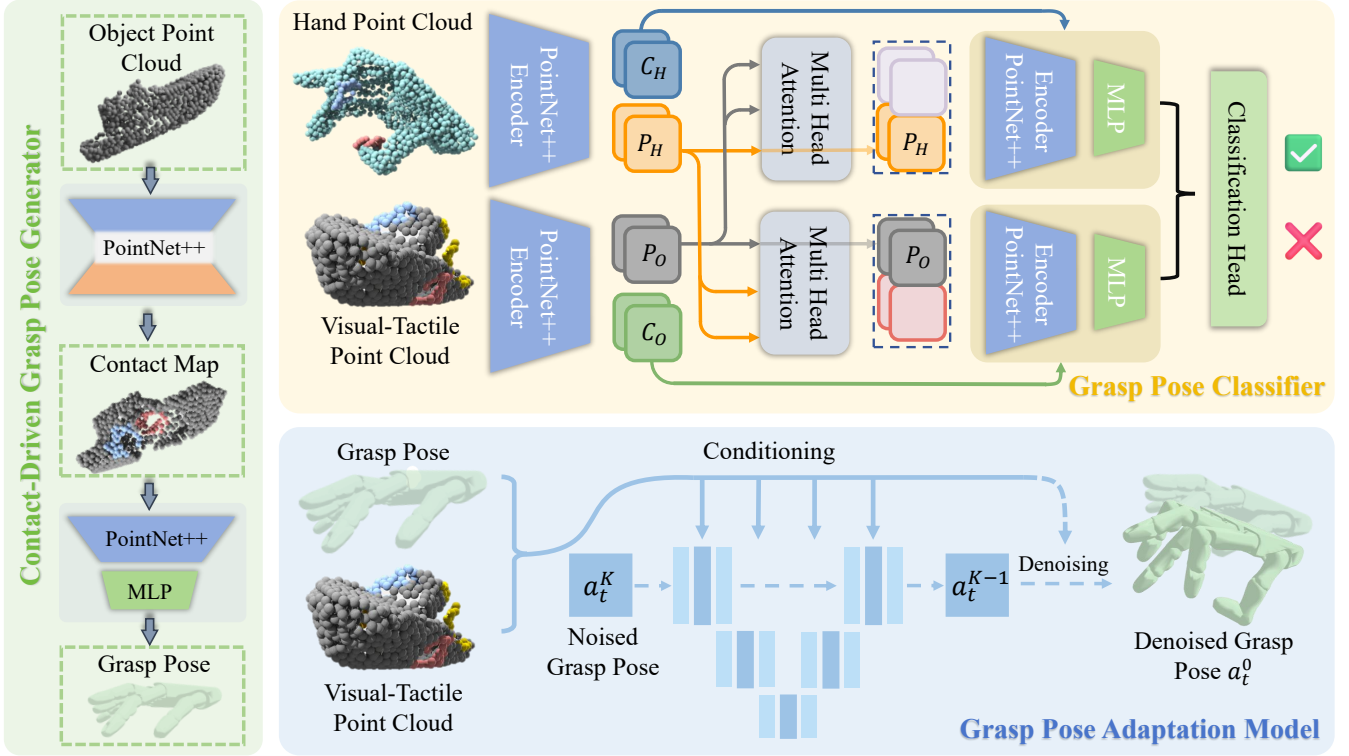


Fig. 3. **Architecture of the grasping framework.** The system consists of a contact-driven grasp generator, a grasp classifier, and a grasp adaptation module for stable dexterous grasping. **Green:** the object point cloud is processed by PointNet++ to predict a visuo-tactile contact map and generate an initial pose. **Yellow:** a dual-encoder PointNet++ architecture encodes the hand and visuo-tactile point clouds, followed by multi-head attention for success prediction. **Blue:** if failure is predicted, a diffusion-based adaptation module iteratively refines the grasp pose conditioned on visuo-tactile features.

The model learns a corrective mapping:

$$\Delta s = g_\phi(\mathcal{P}_{vt}^f, s^f), \quad (8)$$

where Δs represents adjustments to the hand pose and joint angles. The refined grasp is:

$$s^{f, \text{new}} = s^f + \Delta s. \quad (9)$$

The training objective encourages the corrected pose to approach the successful grasp while maximizing predicted grasp stability:

$$\mathcal{L}_{gen} = \|\Delta s - (s^s - s^f)\|_2^2 + \lambda(1 - f_\theta(\mathcal{P}_{vt}^f, s^f + \Delta s)). \quad (10)$$

This objective enables the model to learn corrective actions guided by both tactile perception and grasp success prediction. Figure 4 visualizes the adaptation process.

E. Closed-Loop Grasp Optimization

The contact-driven grasp generator, grasp classifier, and grasp adaptation model are integrated into a closed-loop refinement process. Starting from the initial pose s_0 , the system iteratively evaluates and updates the grasp:

$$\hat{y}_k = f_\theta(\mathcal{P}_{vt}^k, s^{(k)}), \quad (11)$$

$$s_{k+1} = \begin{cases} s_k, & \text{if } \hat{y}_k > \tau_{succ}, \\ s_k + g_\phi(\mathcal{P}_k, s_k), & \text{otherwise,} \end{cases} \quad (12)$$

$$\text{stop if } \hat{y}_k > \tau_{succ} \text{ or } k = K. \quad (13)$$

Here τ_{succ} is success threshold. It terminates when the predicted success probability exceeds τ_{succ} or the maximum iteration number K is reached. The grasp configuration s^* satisfies both geometric feasibility and tactile stability.

III. EXPERIMENTS

A. Settings and Results

We evaluate the proposed visuo-tactile fusion grasping framework in both simulation and real-world settings using a tactile-enabled dexterous hand.

We compare our method against ten state-of-the-art dexterous grasping frameworks, all baselines are evaluated under identical settings using the same grasp success metric.

From Table I, we observe that our method consistently outperforms all baseline approaches across the three object sets, demonstrating superior robustness and generalization.

Furthermore, the performance gap between our method and the baselines increases for unseen categories, indicating that baselines struggle to generalize to novel object shapes and configurations. This emphasizes the importance of incorporating hand-object semantic awareness and closed-loop tactile feedback in developing robust grasping policies. The gains also persist at scale: on the DexGrasp Anything dataset [22] (15k+ objects), our method still reaches 87%.

B. Real-World Experiment

We evaluate our method on more than 20 objects with distinct geometric and material properties. The evaluation

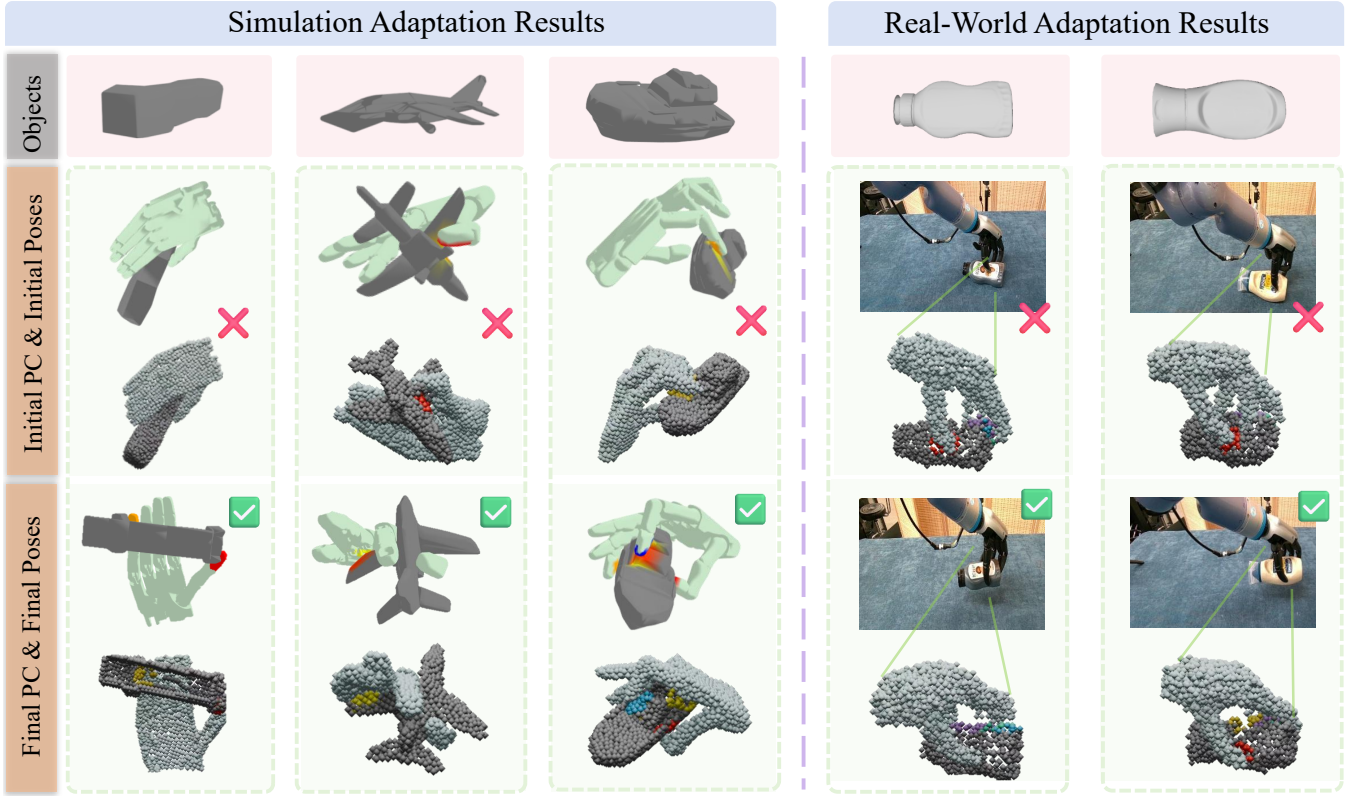


Fig. 4. **Visualization of Dexterous Grasp Pose Adaptation.** Row 1 presents object samples in both simulation and the real world. As shown in Row 2 (hand-object interaction mesh) and Row 3 (hand-object interaction point cloud), the initially generated grasp poses may fail to securely grasp the object, as evidenced by sparse or weak contact regions that lead to loose or unstable grasps. Row 4 and Row 5 illustrate the refined grasp poses and updated contact states after adaptation, where hand-object interaction becomes significantly more stable, with denser contact regions and a larger support area in the point cloud. These observations further validate the effectiveness of the proposed adaptation module in correcting suboptimal initial grasps.

TABLE I
COMPARISON RESULTS BETWEEN BASELINES AND OUR METHOD.

Method	Seen Objects	Unseen Objects	Unseen Categories
UniDexGrasp++ [6]	55%	49%	42%
DexGraspAnything [22]	77%	72%	67%
DexDiffuser [4]	71%	69%	55%
UniDexGrasp (PPO) [23]	52%	46%	40%
Robot Synesthesia	33%	29%	22%
Intuitive Closed-Loop	76%	71%	65%
ContactDexNet [24]	72%	68%	63%
DexGraspVLA [5]	69%	60%	58%
UniDexGrasp-CL	59%	52%	46%
DexDiffuser-VT	75%	71%	57%
Ours	91%	82%	83%

TABLE II
REAL WORLD EVALUATION RESULTS.

Method	Seen Objs	Unseen Objs	Unseen Cats
DexGraspAnything	81%	71%	73%
DexDiffuser	77%	65%	52%
DexGraspVLA	64%	57%	55%
Ours	90%	87%	81%

metrics are consistent with those in simulation. A grasp is considered successful if the object remains stably held for at least three seconds after lifting. Table II reports the final evaluation results. Figure 4 illustrates the hand pose generation and adaptation performance in real world.

IV. CONCLUSION

We presented a unified visuo-tactile-fusion grasping framework that enables robust and generalizable dexterous grasping by tightly coupling tactile feedback with visual geometric perception. Through a contact-aware representation that binds tactile signals with finger identities and a tactile-guided adaptation mechanism for real-time refinement, our method enables fine-grained reasoning over finger-object interactions, enhancing both initial grasp feasibility and adaptive stability in simulation and real-world settings. Experiments demonstrate our superiority in many aspects.

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